

Kenneth Brezinski, PhD, MSc, PEng, CTech, CIT, PMP

Industrial Controls and Network Engineering

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 kbrezinski.github.io

Key Qualifications

- Licensed Professional Engineer (PEng), Certified Technician (CTech), and Chemist In Training (CIT) industrial controls specialist focusing on water and wastewater systems and SCADA/OT networking.
- Hands-on experience with IO mapping, IP subnetting, loop drawings, graphic standards (ISA 10.1), process control narratives, alarm management (ISA 18.2), FAT/SAT, control cabinet/kiosk design; able to design in AutoCAD Electrical with minimal supervision.
- Experience in designing and troubleshooting Allen Bradley (1769 and 5069) and Schneider (M340, M580 and SCADAPack) PLCs; SCADA development with VTScada and Ignition 8.1.
- Advanced knowledge of networking concepts (Network+ and Security+ certified) for IT and OT network deployments and cybersecurity assessments (per IEC 62443), including intrusion prevention and endpoint products (mainly Cisco, Fortinet and Palo Alto; EDR for Microsoft).
- PhD graduate and Post Doctoral Fellow in applied deep learning for enterprise malware threat detection, with experience programming big data, and large scalable graph, transformer, image and DNN frameworks; authorized over a dozen publications in Cyber Security and Artificial Intelligence.

Work Experience

Intermediate Instrumentation and Controls Engineer

11/24-present

WSP Canada, Vancouver, Canada

- Work with the Vancouver Water & Wastewater Industrial Controls and Automation Group on developing electrical and controls design services for municipal, rural and Metro Vancouver clients, with project fees in excess of 1m+.
- Design requirements and equipment selection for control panels (120-600VAC), including main and marshalling panel enclosures, UPS (inc. bypass), ATS/MTS, power supplies, breaker sizing, networking (firewalls, switches), fuses, terminal blocks, relays; grounding, cable scheduling.
- Commission Firewalls, Switches, Protocol Converters, Access Points, Server Hardware, Virtual Machines for SCADA deployments (VTSCADA and Ignition), including VLAN segregation and subnetting, over a variety of protocols (Modbus TCP, Ethernet/IP, Fieldbus, Profibus, RS-485)
- Equipment selection and single line design (4-20mA and 0-10V 2- and 4-wire loops) for pumps, flow, pressure, temperature, transmitters, building alarms; VFD and MCC selection and requirements.
- Participate and lead SCADA workshops, PCN discussions; author SCADA masterplans; Front End MMCD, Division 1, Division 13 and 16 Specifications (5- and 6-digit).

Systems Designer, Water & Wastewater

05/23-11/24

WSP Canada, Winnipeg, Canada

- Work with the Water & Wastewater Treatment group on water and wastewater solutions for the prairie region of Canada.
- Carrying out water treatment plant assessments, investigations, and process design & system optimization; including carrying out aqueous modelling of complex systems (Kow, Ksp, kinetics) in the water, wastewater, mining, and oil and gas sector using Geochemist Workbench.
- Lead proposal efforts and respond to RFQ and RFPs for new contracts with industrial and municipal clients.

Post Doctoral Fellow, *University of Manitoba*, Winnipeg, MB

11/24-present

- Lead research efforts in close collaboration with Canadian Tire Corp. executives with a focus on Malware detection of enterprise security threats and complexity analysis applied to machine learning and deep learning frameworks.
- Research agentic large language models for automated code generation for IEC 61131-3 compliant code that meets SIL requirements as per IEC 61508 for water and wastewater systems.

Visiting Researcher, *National Institute for Informatics*, Tokyo, Japan

09/22-12/22

- Develop a graph autoencoder to detect network anomalies from backbone network traffic connecting Japanese Academic institutions to North America.

Data Scientist Intern, *Microsoft*, Redmond, WA

05/22-08/22

- Worked with the Windows Defender for Endpoint Team on developing a supervised ML detector to alert customers in the early stages of an exfiltration or ransomware attack

Applied Research Scientist II Intern, *Amazon Web Services*, New York, NY

05/21-08/21

- Worked with the Amazon GuardDuty threat detection research team on developing a novel meta-learner to apply weak labels to Linux binaries

Technical Skills

PLC / SCADA	RSLogix 500, Designer 5000, SCADAPack workbench, AB Architecture Builder, Ignition 8, ControlExpert
Electrical	AutoCAD Electrical, ETap
Programming	Python (Pytorch, Tensorflow, Geometric PySpark, Git, JAX, Flax, AWS-EMR), Java, Matlab, LaTeX, MLOps (Docker, Streamlit, Plotly, MLFlow, Kinesis)
Malware Analysis	Static analysis tools such as PE View, Bintext, Dependency Walker, PEiD, OllyDBG, IDAPro; Dynamic tools such as Procmon, BurpSuite, Wireshark, API monitor; Splunk

Professional Memberships

EGBC	Professional Engineer (PEng) , Engineers and Geoscientists BC
EngGeoMB	Professional Engineer (PEng) , Engineers Geoscientists Manitoba
ISA	Member , Industrial Society for Automation
ASTTBC	Certified Technician (C.Tech.) , Applied Science Technologists & Technicians of BC
CTTAM	Certified Technician (C.Tech.) , Certified Technicians & Tech. Association of MB
PMI	Project Management Professional (PMP) , Project Management Institute
CompTIA	Security+ , Computing Technology Industry Association
CompTIA	Network+ , Computing Technology Industry Association
ACPA	Chemist in Training (C.I.T.) , Association of the Chemical Profession of Alberta
WCW	Member , Western Canada Water
WEF	Member , Water Environment Foundation

Publications

A full list of my academic publications can be found on my [personal site](#).

 Education

- 08/18-08-24 **Doctor of Philosophy**, *University of Manitoba*, Winnipeg, MB.
Electrical and Computer Engineering
- 01/16-09/18 **Master of Science**, *University of Manitoba*, Winnipeg, MB.
Civil Engineering
- 08/10-08/15 **Bachelor of Science**, *University of Winnipeg*, Winnipeg, MB.
Chemistry